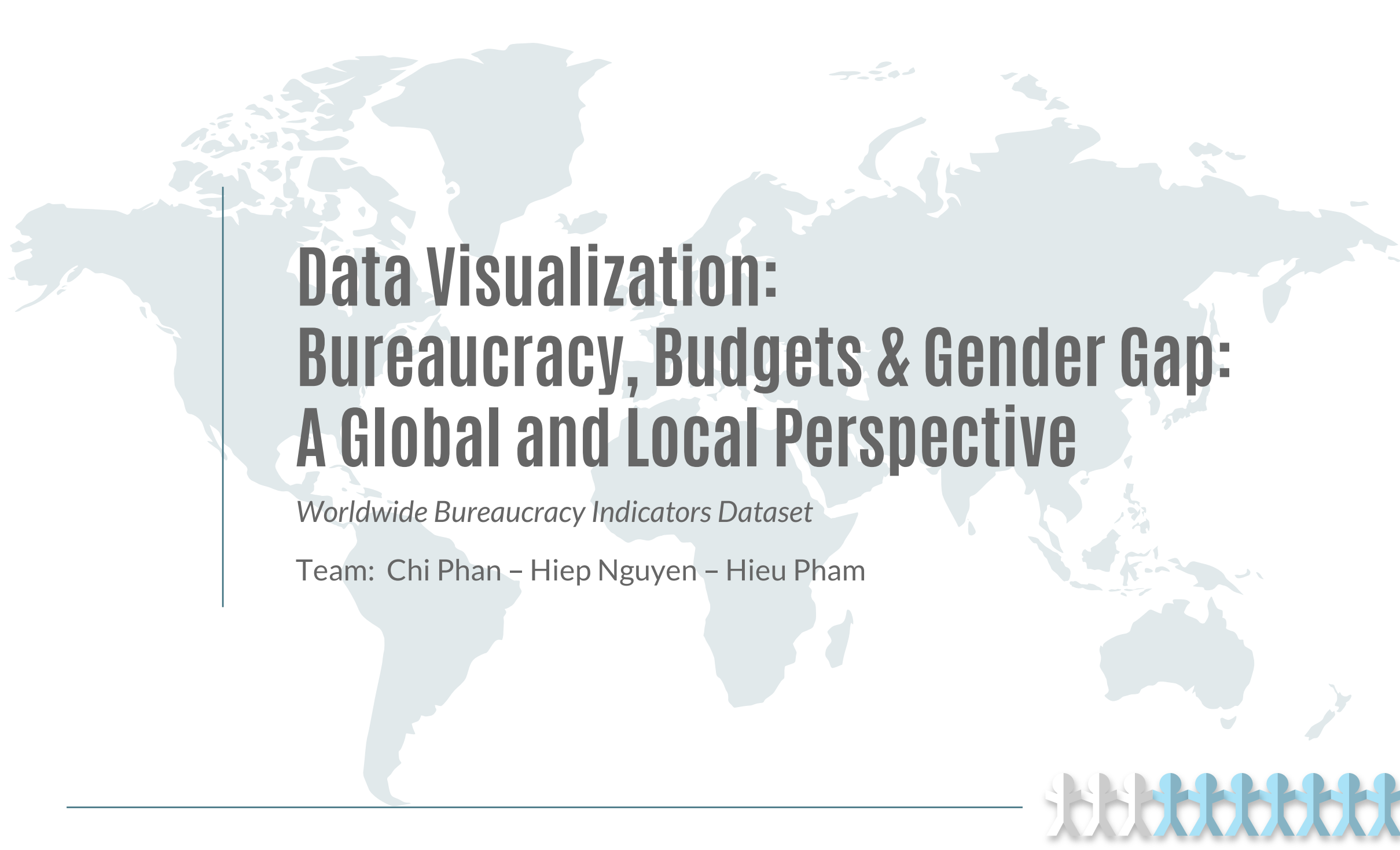




Data Visualization: Bureaucracy, Budgets & Gender Gap: A Global and Local Perspective

Worldwide Bureaucracy Indicators Dataset

Team: Chi Phan – Hiep Nguyen – Hieu Pham



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DATASET INTRODUCTION:

Worldwide Bureaucracy Indicators (WWBI)

Dataset:

- A cross-national dataset on public sector employment and wages developed by the **World Bank's Bureaucracy Lab**.
 - Based on 53M unique survey observations and consist of **192 indicators** for **202 countries** and territories **between 2000 and 2020**.
 - Encompasses **3 types of variables**: characteristics of public-sector employment, wages & compensations, and the overall wage bill
- ⇒ Provide a holistic picture of the structure, equity, and evolution of public sector employment across countries and over time

Data Source: World Bank

Data Dictionary: TidyTuesday adaptation

wwbi_data.csv

variable	class	description
country_code	character	3-letter ISO_3166-1 code
indicator_code	character	code identifying the indicator of bureaucracy
year	numeric	year of the data
value	numeric	numeric value of the data

wwbi_series.csv

variable	class	description
indicator_code	character	code identifying the indicator of bureaucracy
indicator_name	character	name of the indicator

wwbi_country.csv

variable	class	description
country_code	character	3-letter ISO_3166-1 code
short_name	character	short or common name for the country
table_name	character	more alphabetically sortable name of the country
long_name	character	full name of the country
x2_alpha_code	character	2-letter ISO_3166-1 code
currency_unit	character	currency unit
special_notes	character	special notes
region	character	region
income_group	character	low, lower middle, upper middle, or high income



INTRODUCTION - OUR OBJECTIVES



Understanding Government Spending Priorities Worldwide

Investigate how countries allocate public wages & gain insight into their economic focus and governance.



Examining Gender Equity in the Workforce

Uncover structural differences in gender representation & understand how different systems influence workforce composition.



Comparing Public and Private Sector Dynamics

Discover systemic differences that affect fairness, efficiency, and representation.



QUESTION 1 - Global Trends In Public Wage

“How has the proportion of GDP allocated to public wages changed over time across different world regions and income groups?”



Why this matters?

- Plays as a **crucial indicator** of how governments manage **fiscal resources**
- Reveals important **cross-country differences** in public sector management and state capacity
- Partially informs decisions about reforming **public compensation systems** and prioritizing **fiscal adjustments** to support sustainable growth



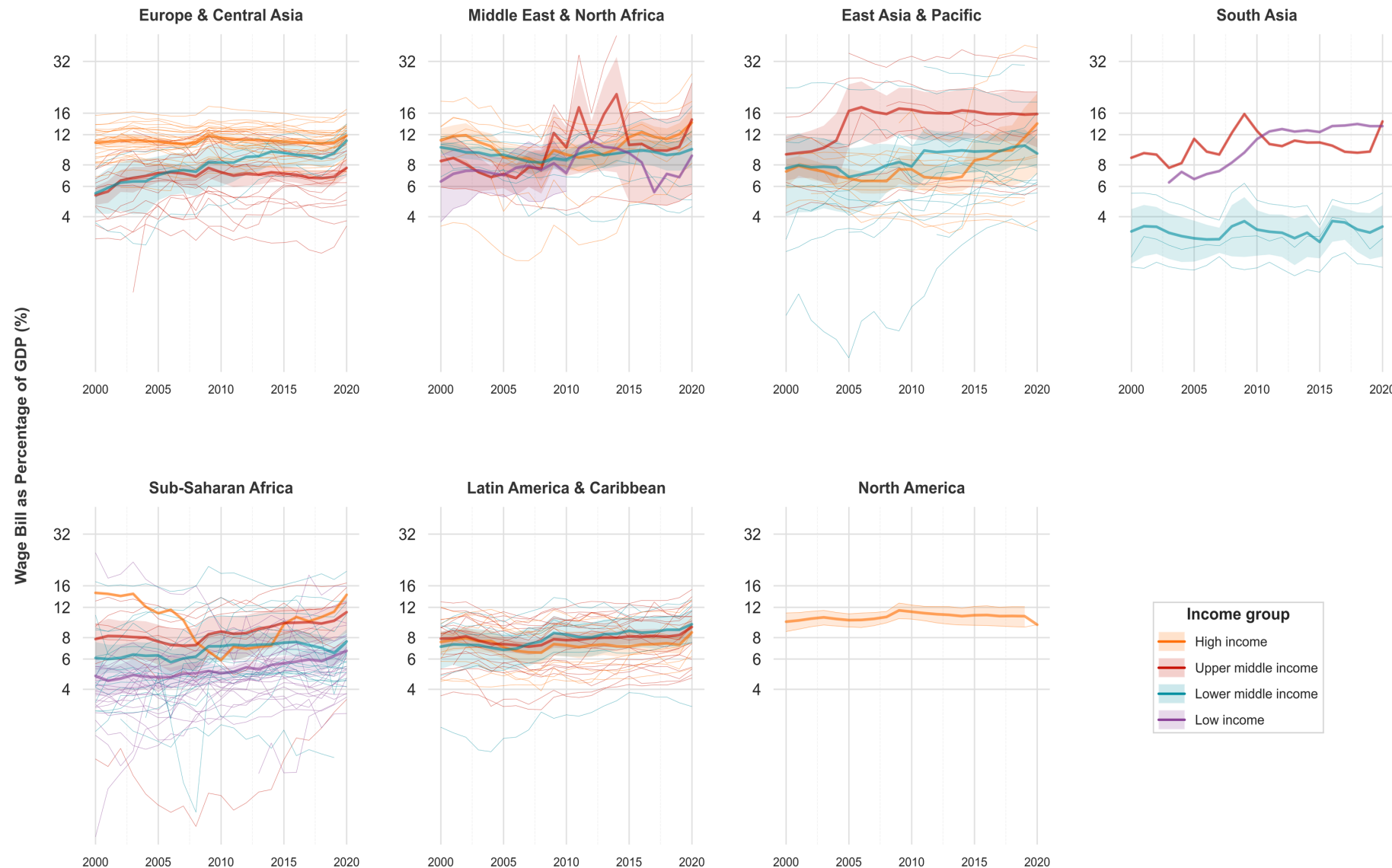
Data & Method

- **6 variables included:** *country_code, region, income_group, indicator_code, year, value*
- **Handling missing data:** drop countries with unknown regions/income groups.
- **Aggregation:** seaborn smooth for country-level noise.
- **Preprocessing:** apply a log scale to emphasize regions' differences.
- **Visualization:** Use line charts with facets by region and colors by income group.



QUESTION 1 - Visualization & Findings

Wage Bill as Percentage of GDP by Region and Income Group



“How has the proportion of GDP allocated to public wages changed over time across different world regions and income groups?”

Insights:

- While **Latin America & Caribbean** maintains the most stable wage levels (std=2.68%), **East Asia & Pacific** has experienced the most dramatic changes (std=7.75%).
- **High-income groups** consistently maintain stable wage bills (8-15%), while **low-income groups** typically have the lowest percentages (often below 8%)
- The interesting fluctuation in Middle East & North Africa

QUESTION 2: Female Representation In Occupation Groups

How has the gap in female representation between high-level (managers, professionals, technicians) and low-level (clerks, elementary workers) occupations evolved over time in Vietnam, and how does it differ between the public and private sectors?

? Why this matters?

- Gender gaps persist in career progression, not just access
- Public and private sectors may treat women differently
- Informs policy for leadership diversity and inclusion

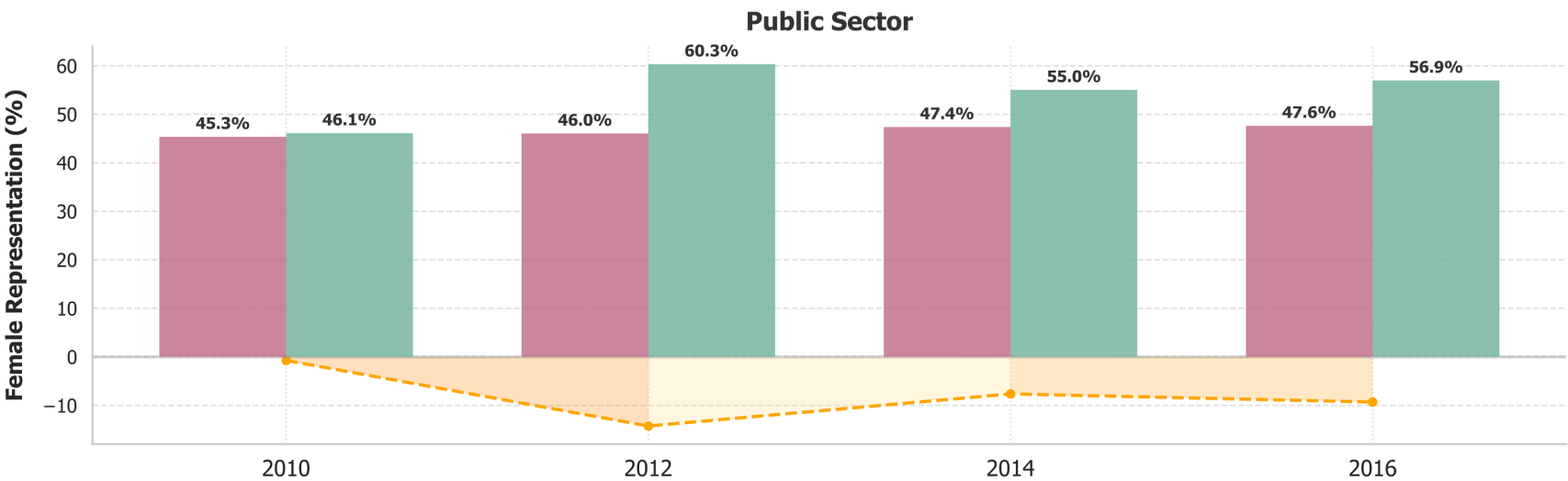
Data & Method

- Filtered Vietnam and gender indicators from cleaned dataset
- Aggregated average female share in each role group
- Calculated gap = % high-level – % low-level for each year and sector
- Visualization: Combine grouped bar chart + area chart



QUESTION 2 - Visualization & Findings

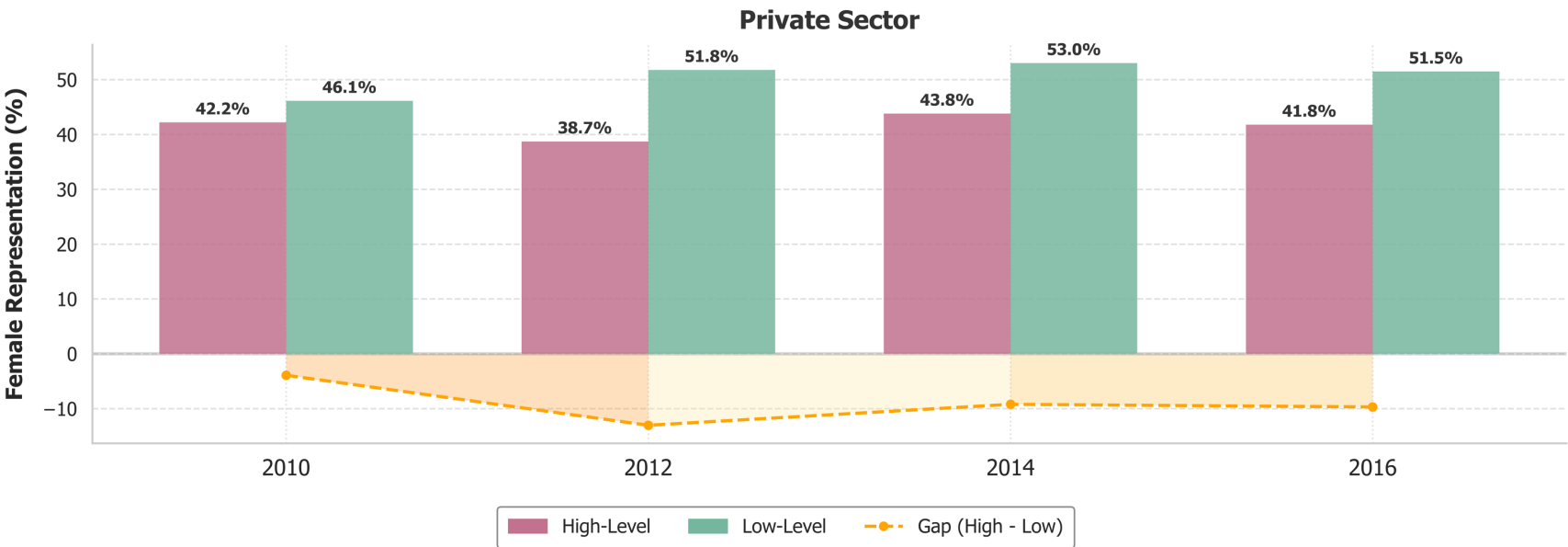
Female Representation by Occupation Level in Vietnam (2010-2016)



“How has the gap in female representation between high-level and low-level occupations evolved over time in Vietnam, and how does it differ between the public and private sectors?”

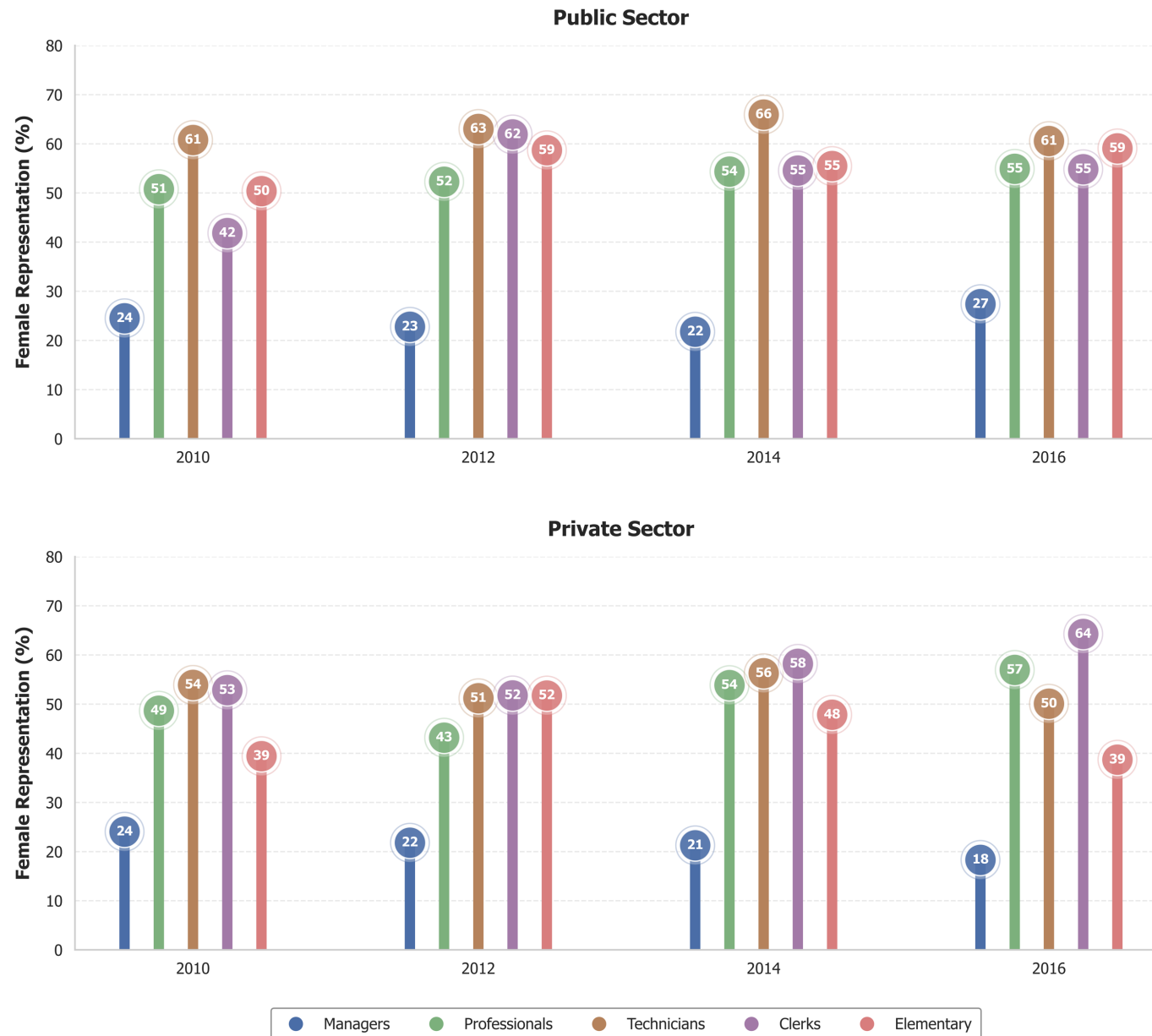
Insights:

- **Women hold more low-level than high-level jobs** in both sectors throughout 2010–2016.
- **Private sector shows a bigger gender gap** compared to the public sector
- Although the public sector shows a modest improvement in high-level representation, **the overall gap persists over time** in both sectors.



QUESTION 2 - Visualization & Findings

Female Representation by Occupation Type in Vietnam (2010-2016)



“How has the gap in female representation between high-level and low-level occupations evolved over time in Vietnam, and how does it differ between the public and private sectors?”

Insights:

- **Manager** roles have the **lowest** female share.
- **Clerks dominate** the female workforce, especially in the private sector.
- The public sector is **more balanced** across roles.

SUMMARY & FUTURE DIRECTION

Finding Summary & Discussion

- Public wage spending **varies greatly by region and income level**. High-income regions maintain consistent wage bills (8–15%), while low-income countries tend to allocate less (often <8%).
- **Vietnam's workforce shows persistent gender gaps**, with women underrepresented in higher-level jobs (managers, professionals) in both public and private sectors.

Limitations

- **Missing or incomplete data in some countries or time periods**, especially for early 2000s and developing countries.
- **Vietnam analysis limited to only 4 years** (2010, 2012, 2014, 2016), due to data availability tied to official survey coverage and base-year constraints.

Future Direction

- **Extend the time frame or update with recent data** for post-2016 trends for Vietnam
- **Incorporate contextual economic and policy data** (e.g., GDP growth, education, gender equality indices) to enrich interpretation.
- **Compare more countries** in Southeast Asia or similar economies to draw regional insights beyond Vietnam.



THANKS!

Do you have any questions?

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